Additional file 1

How accurate is altitude from pressure geolocators? Supplementary methods and results

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This supplement describes the selection of reference stations (S1), environmental variation (S2), the ERA5-Land comparison (S3), the bird flight-height distribution (S4), and horizontal-position sensitivity (S5). It uses the main manuscript's conversions and error definitions. Code and underlying numerical summaries accompany the analysis [1].

S1 Reference-station selection

We select reference stations whose pressure records and reference heights can be trusted, spread evenly across the globe. Both datasets follow the same sampling design:

- **One station per cell.** We divide the globe into 200 × 200 km equal-area cells (Equal Earth projection) and select at most one station per cell.
- **Random candidate order.** Within each cell, candidates are assessed in random order until one passes every check.
- **Latest qualifying year.** For each candidate, we use the most recent 12-month period between 1 January 2022 and 12 October 2026 that passes every check.

S1.1 GHCNh ground stations

We use station pressure and reported elevation from NOAA's Global Historical Climatology Network hourly archive (GHCNh) [2]. Every station with coordinates, an elevation and pressure observations in the study period is a candidate, giving candidates in 2,962 cells. We assessed 6,073 candidates before each cell either had a qualifying station or no candidates left. Each candidate is rejected at the first of the following steps it fails; the number rejected is given with each step. We set the thresholds by visual inspection of pressure time series.

1. **Check the reported elevation.** The reported elevation is the reference against which converted altitudes are compared, so it must describe the station's actual site.

    - *Terrain.* The reported elevation must match the terrain within 50 m, allowing for the range of terrain elevations within 500 m of the station's coordinates. We use the 30 m Shuttle Radar Topography Mission (SRTM) v3 [3] elevation model, with Advanced Spaceborne Thermal Emission and Reflection Radiometer Global Digital Elevation Model (ASTER GDEM) v3 [4] where SRTM has no data. Stations without terrain data, mostly on reefs and in Antarctica, are also rejected (330 candidates).
    - *Documented conflicts.* We exclude stations whose metadata show that the reported elevation is unreliable, for instance drifting ice stations, or coordinates that identify a different site from the station (59 candidates).

2. **Prepare hourly pressure.**

    - We keep station pressure measurements between 300 and 1100 hPa that the archive's quality codes [5] do not mark as suspect, erroneous, removed or failing quality checks.
    - Where an hour has several values, we keep the one closest to the full hour.
    - We remove isolated spikes: single values that depart from both neighbours, which agree with each other. A spike departs by more than 5 hPa from neighbours within 3 h that agree within 2 hPa or, in sparser records, by more than 10 hPa and at more than 2 hPa per hour from neighbours within 12 h that agree within 3 hPa.

3. **Require enough data.** A 12-month period qualifies only if each of its months has at least 15 days with at least four hourly pressures (3,025 candidates have no such period).
4. **Reject large jumps.** A period fails if it contains a pressure change of more than 20 hPa within 3 h, or of more than 50 hPa across a data gap of up to 7 days (9 candidates).
5. **Check the reference height and timestamps.** A smooth pressure series can still refer to the wrong height or time. Many stations also report sea-level pressure or an altimeter setting, which they calculate from station pressure and the barometer height they use. Inverting these values recovers that height from the station's own reports.

    - *Implied height.* The altimeter setting follows a standard-atmosphere formula and inverts exactly, so its implied height must agree with the reported elevation within 15 m. Sea-level pressure is inverted using the reported air temperature and a lapse rate of 6.5 K km^{-1}. National reduction conventions differ, so its tolerance is 15 m plus 1% of elevation, and it is only used when its implied height is stable over the period. The altimeter setting is used where available. A candidate is rejected if:
        - no period has at least 100 implied heights in every month, so its reference cannot be verified (585 candidates);

  - the median implied height departs from the reported elevation by more than the tolerance in any month of the period, by more than three times the tolerance on any day, or by more than the tolerance over the station's full record or in 12 or more consecutive months of that record (231 candidates). The daily limit catches days on which the station pressure itself is offset.
- *Timestamps.* The semi-diurnal atmospheric tide peaks near 09:48 local solar time worldwide. A period fails if this tide, fitted to pressure departures from daily means, has an amplitude above 0.5 hPa and peaks more than 1.5 h away from 09:48, which indicates timestamps not referenced to UTC (21 candidates).

6. **Visual review.** We inspected the pressure series of the selected stations with the largest ERA5 errors and excluded 19 with a defect that the automatic checks miss (Table 1). Their cells were refilled from the remaining candidates.

The final sample contains 1,794 stations, and spike removal discards 1,120 values from their selected periods. The checks can miss sensor offsets shared by station and sea-level pressure, timestamp errors where the tide is weak, and faults that leave a plausible pressure series. Median summaries limit the influence of any remaining extreme errors.

Table 1: GHCNh stations excluded after visual review of their pressure series.

Station	Name	Defect
UZM00038403	Buzaubaj	Slow pressure drift of up to about 15 hPa over several weeks
BLU085109-1	San Ramon	Gradual pressure drift of about 5 hPa over the year
FSU98404001	Kerguelen	Large irregular departures consistent with inconsistent report timing
FSU98404004	Crozet	Large irregular departures consistent with inconsistent report timing
FSU98404002	Nouvelle Amsterdam	Large irregular departures consistent with inconsistent report timing
USW00063819	Boone Watauga Co Hsp	Persistent pressure step of about 5 hPa
INMU0020523	Honavar	Recurring reports offset by about 4 hPa from the main pressure level
BRM00081835	Apodi	Reports alternating between two pressure levels about 8 hPa apart
SFM00068481	Pongola	Persistent pressure step of about 3 hPa
IDU097570-1	Serui	Persistent pressure step of about 1.5 hPa
TXM00038388	Ekezhe	Pressure alternating between two levels about 4 hPa apart over multi-day blocks
KZM00036806	Druzba	Irregular multi-day pressure departures of up to 6 hPa
THM00048302	Doi Ang Khang	Recurring multi-day pressure excursions of about 5 hPa
TXM00038507	Turkmenbashi	Successive pressure steps of 1 to 2 hPa
RPM00098630	Cuyo	Persistent pressure step of about 3.5 hPa
BRM00086995	Rio Grande	Persistent pressure step of about 2.5 hPa across a reporting gap
AVM00078846	Anguilla	Repeated pressure steps of 1 to 1.5 hPa
IRM00040832	Abamusa Island	Irregular multi-day pressure departures of up to 8 hPa
BRM00081909	Campos Sales	Months of large irregular scatter followed by a drift of about 5 hPa

S1.2 IGRA2 soundings

We use pressure, temperature and reported geopotential-height profiles from NOAA's Integrated Global Radiosonde Archive (IGRA2) [6–8]. Each sounding includes a ground-level record at launch, which provides the surface reference for flight-height errors. We assessed 828 candidate stations, each rejected at the first of the following steps it fails.

1. **Prepare sounding profiles.**

    - Each sounding needs a valid surface record. We prepare pressure–height pairs up to 6 km above the launch surface, so that the profile extends beyond the 5 km evaluated in the analysis. Missing surface heights take the station elevation.

- The time of each level is the launch time plus its reported ascent time. Missing launch times take the nominal launch time, and missing ascent times assume 5 m s^{-1}, a typical radiosonde ascent rate.

2. **Check each sounding.** The archive's heights are not measured directly: they are calculated from the sonde's pressure and temperature with the hypsometric equation. Recalculating each level's height above the surface from the reported temperatures should therefore reproduce the reported height; a mismatch reveals an erroneous surface record, height reference or reported height.

   - We average temperature over log pressure from the surface to each level, extending the lowest temperature to the surface and interpolating missing values.
   - A sounding is consistent if every level agrees within 30 m plus 3% of its height above the surface. This margin covers the omitted humidity effect and coarse temperature profiles.
   - Inconsistent soundings are discarded, as are soundings with fewer than two temperatures.

3. **Find the latest qualifying year.** A 12-month period qualifies if each of its months has at least 15 launch dates with a consistent sounding (185 candidates have no such period), and if no more than 10% of its checked soundings are inconsistent (14 candidates). We retain all levels of the consistent soundings in that period.

The final sample contains 629 stations.

S2 How the error varies

Ground: elevation gap, climate and time

We assign stations to the five main Köppen–Geiger climate zones from their coordinates using kgc [9]. Stations without a classification are omitted from climate-specific summaries. We compare errors by season and local solar time, shifting southern-hemisphere seasons by six months. We model absolute station bias and SD against the absolute station–ERA5 elevation difference using generalized additive models.

The station–ERA5 elevation gap is associated with within-station scatter (Figure 1): fitted SD rises from 3.3 m at zero gap to 10.0 m at a 1000 m gap. Ground errors vary little with season and local time (Figure 2); among climate zones, polar stations have the largest within-station scatter (Figure 3).

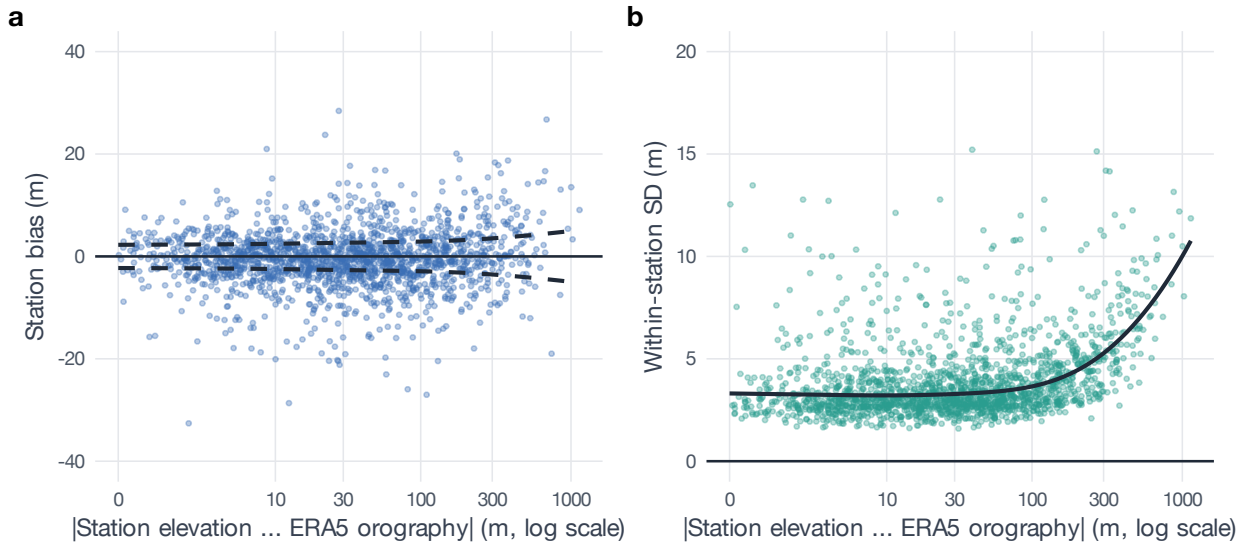


Figure 1: Station bias (a) and within-station SD (b) against the absolute difference between station elevation and ERA5 orography, ERA5 standard conversion. Curves are back-transformed generalized additive model fits of log(|bias| + 0.5 m) (dashed, shown as ±) and log(SD) (solid) on log(1 + elevation difference).

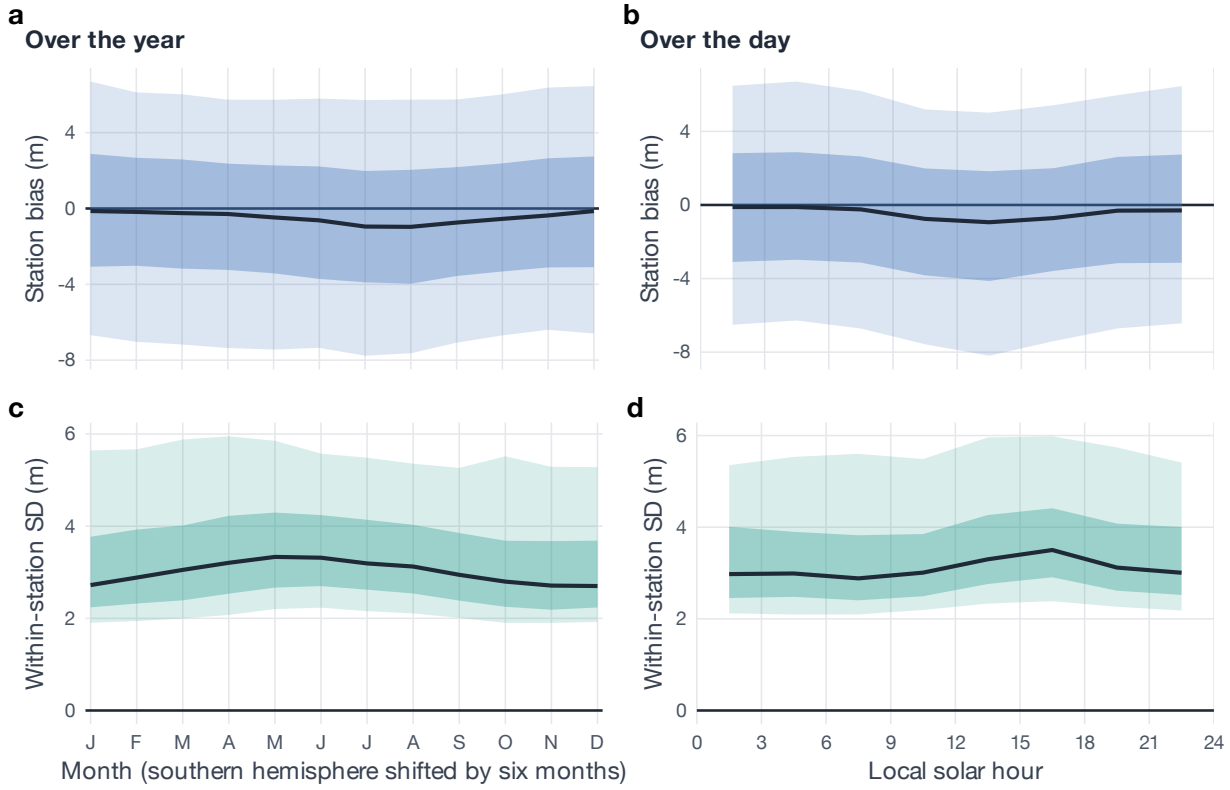


Figure 2: Station bias (a, b) and within-station SD (c, d) by month (southern hemisphere shifted by six months) and 3 h local-solar-time band, ERA5 standard conversion. Lines: medians across stations; bands: 25th–75th and 10th–90th percentiles across stations. Station classes with fewer than 30 observations are omitted.

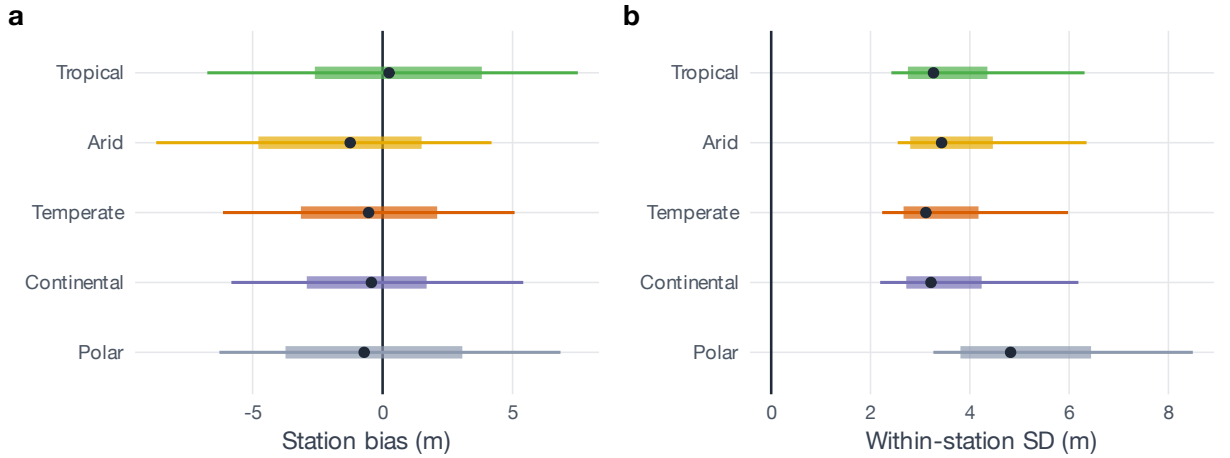


Figure 3: Ground errors by main Köppen–Geiger climate zone, ERA5 standard conversion: (a) station bias and (b) within-station SD. Points: medians across stations; thick and thin segments: 25th–75th and 10th–90th percentiles.

Flight: column temperature

To test whether flight-height errors arise from the assumed temperature profile, we compare each observed error with the error predicted from the difference between the assumed and the radiosonde-derived mean column temperature. For ERA5 with the standard lapse rate, the predicted from the assumed versus observed mean column temperature closely follows the surface-referenced error ($r = 0.98$; Figure 4). Most of the systematic flight-height error therefore arises from the assumed temperature profile; its variation with climate, season and time of day is shown in the main manuscript.

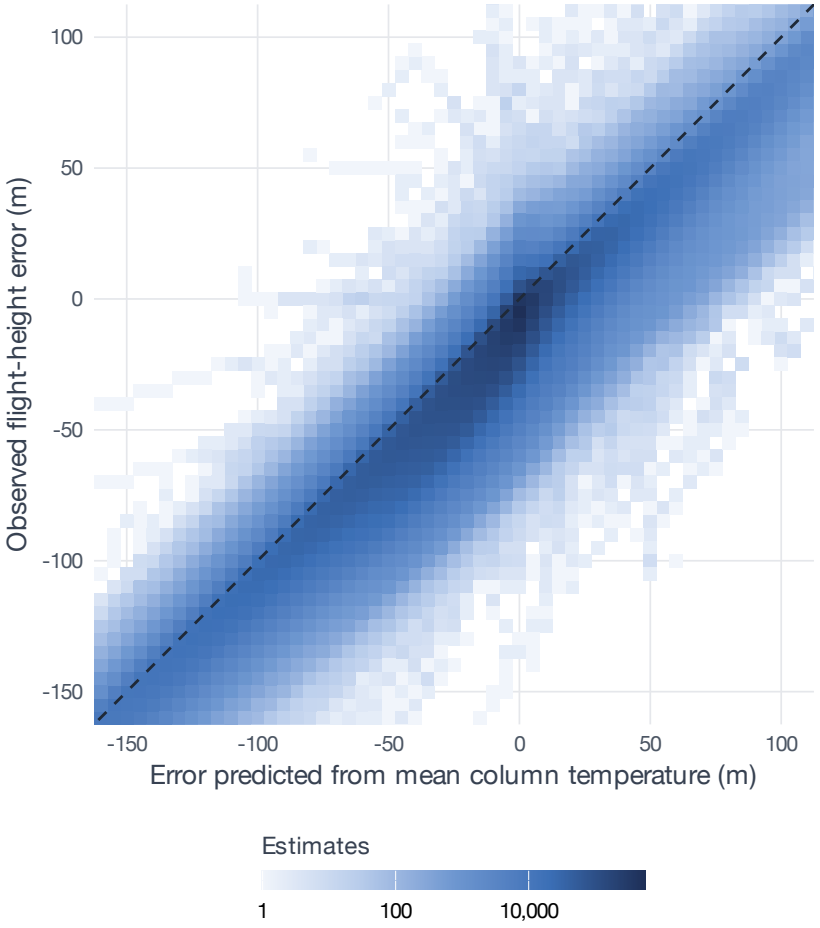


Figure 4: Observed flight-height error against the error predicted from the difference between assumed and observed mean column temperature, ERA5 standard conversion, at heights more than 200 m above the sounding surface. Tile colour: number of estimates (log scale); dashed line: 1:1. The display is limited to −150 to 100 m.

S3 ERA5-Land as surface reference

ERA5-Land provides surface conditions on a finer 0.1° grid than ERA5 single levels [10]. We replace the ERA5 single-level surface pressure, temperature and elevation with their ERA5-Land equivalents

in the standard-lapse-rate conversion and compare both products at the GHCNh stations over the hours with inputs from both. ERA5-Land produces larger errors in most hours and at most stations (Figure 5). The increase is largest for the persistent offset, which indicates a difference in the surface reference rather than in the representation of temporal variability.

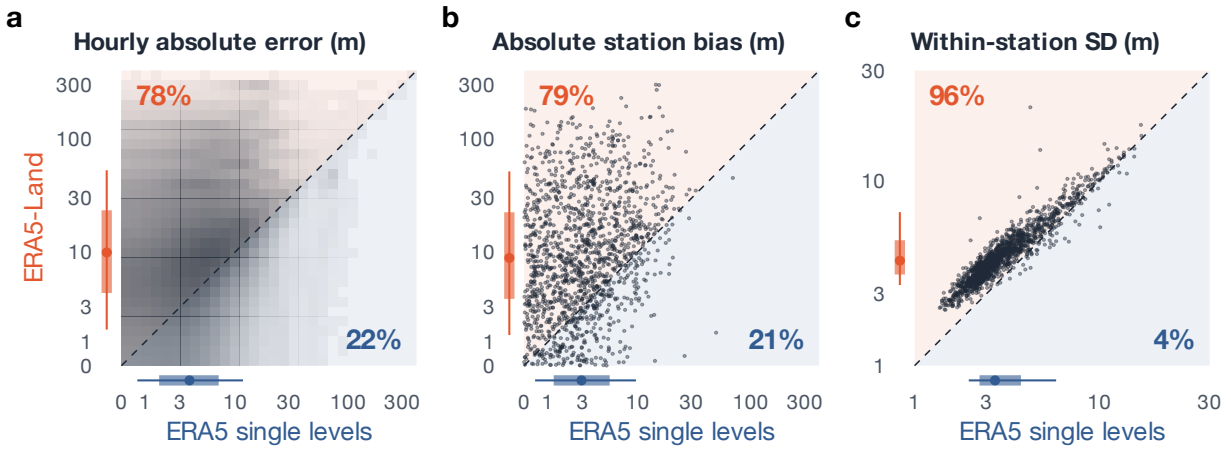


Figure 5: Ground errors with ERA5 single-level (horizontal axis) and ERA5-Land (vertical axis) surface conditions, standard lapse rate, at the same stations and hours: (a) hourly absolute error (shading: density of hours, equal total weight per station), (b) absolute station bias and (c) within-station SD (points: stations). Percentages give the share of hours or stations above (ERA5-Land larger) and below the 1:1 line. Axis bars: median, 25th–75th (thick) and 10th–90th percentiles (thin). Axes are pseudo-logarithmic.

S4 Bird flight-height distribution

We estimate the flight-time height distribution from 203 tagged birds of 25 species in the Master GeoLocator Data Package [11]. Flights have been manually classified using activity measurements and pressure changes. The aim is to approximate how much flight time birds spend at different heights above ground, providing ecological context for the height-dependent conversion errors.

We use the altitude above sea level supplied with each reconstructed trajectory. Missing altitudes are calculated from tag pressure and ERA5 surface pressure, temperature and elevation using the standard lapse rate. Ground elevation is the median Copernicus GLO-90 terrain elevation within 0.25° cells along the reconstructed most-likely path, obtained using GeoPressureR's `path2elevation()` [12]. We interpolate this terrain profile to pressure observations assuming constant speed between successive stationary locations, then subtract it from sea-level altitude. Figure 6 illustrates the conversion over one flight and shows the resulting flight-time distributions. Heights remain approximate because positions are uncertain and terrain varies within each cell.

Each observation contributes the time until the next record, capped at one hour. Negative height estimates are counted in the lowest height band.

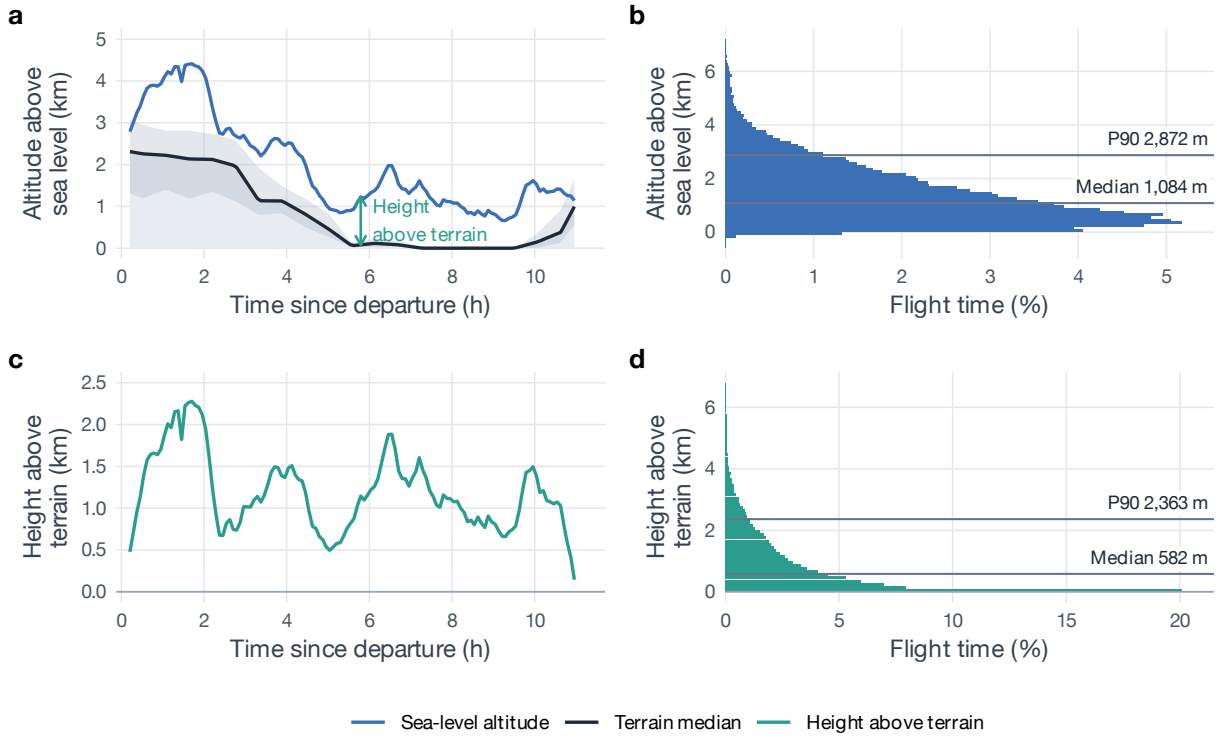


Figure 6: From sea-level altitude to flight height above terrain. (a, c) One flight of a ring ouzel (tag 16LF, 10–11 October 2017): (a) sea-level altitude and median terrain elevation, with the 10th–90th percentile range of terrain within each cell shaded, and (c) the resulting height above median terrain. (b, d) Flight-time distributions of the same two quantities for all tracked birds in 100 m bins, with medians and 90th percentiles (P90).

S5 Sensitivity to horizontal position

Geolocator positions are uncertain, so the atmospheric conditions used in the conversion are taken at an approximate location. To estimate how much this matters, we displace the location at which ERA5 and GPT3 are queried while keeping the measured pressure, the time and the radiosonde reference height unchanged. The standard atmosphere does not depend on location and is unaffected.

This pilot uses ten IGRA2 stations spread across the five main climate zones, with flat and rough terrain (sub-grid orography SD below or above 100 m) and both hemispheres. For each station, we take two launches per quarter of 2024 and compare the level nearest 2 km above the surface (within 1.7–2.3 km) and the launch surface record. We displace the query by 10, 25, 50, 100 and 200 km in eight directions and compare the resulting errors with those at the exact location.

Displacement changes the error of individual estimates more than the overall error distribution (Figure 7). With the corrected ERA5 conversion, a 50 km displacement changed the median absolute error of absolute altitude by 0.2 m and that of surface-referenced flight height by -1.1 m, while individual estimates changed by a median of 5.1 m (95th percentile 21 m). At 200 km, the median absolute error of absolute altitude increased by 4.2 m and individual estimates changed by a median

of 12.5 m. The pilot covers a single height and a small station sample, and does not include uncertainty in the terrain beneath a bird.

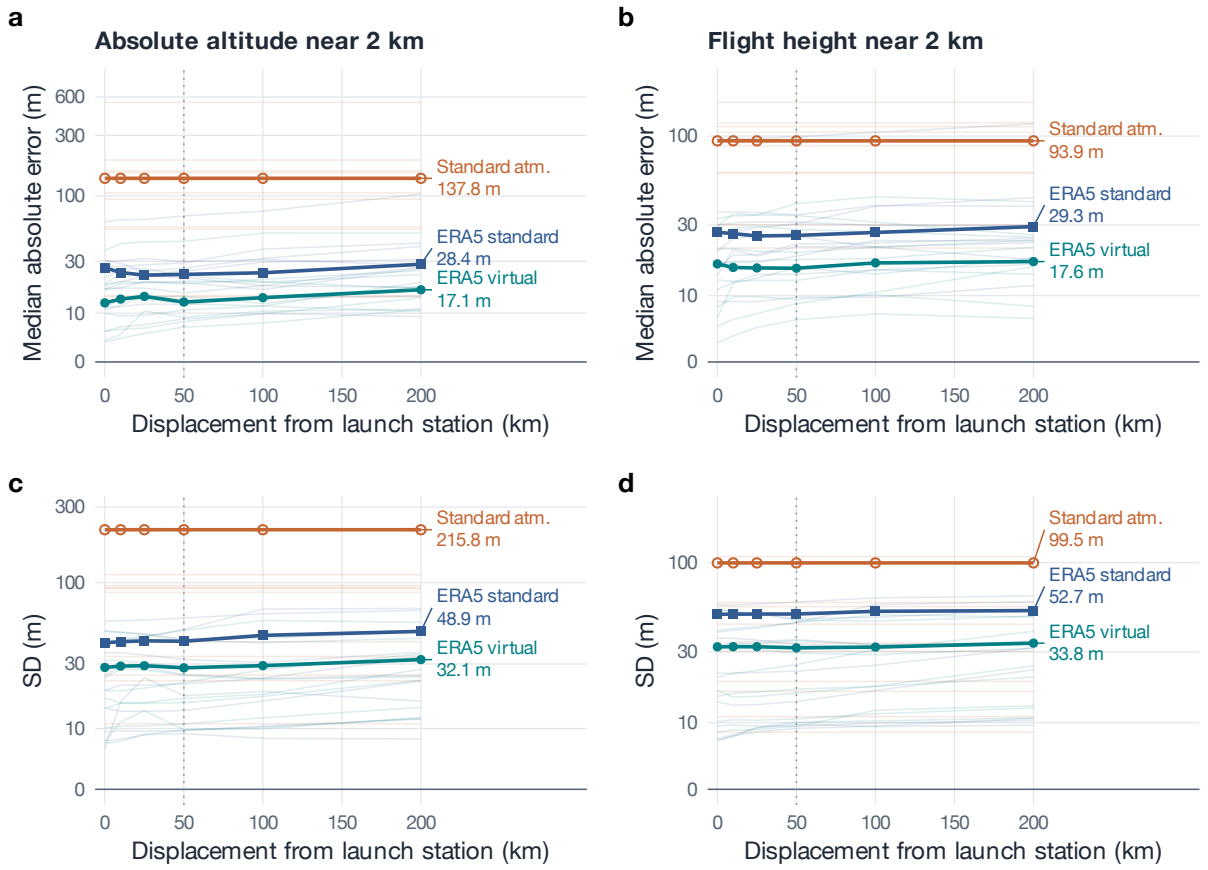


Figure 7: Errors of the three conversions against the displacement of the atmospheric query from the launch station, for absolute altitude near 2 km (a, c) and surface-referenced flight height (b, d): median absolute error (a, b) and pooled SD (c, d). Faint lines: individual stations; bold lines: all stations pooled. The standard atmosphere does not depend on location. Vertical axes are pseudo-logarithmic.

References

1. Nussbaumer R, Gravey M. How accurate is altitude from pressure geolocators? Code, data and report. 2026. https://doi.org/10.5281/zenodo.23085093.

2. NOAA National Centers for Environmental Information. Global historical climatology network hourly (GHCNh). 2026.

3. Open Topo Data. SRTM: Dataset and public API documentation.

4. Open Topo Data. ASTER: Dataset and public API documentation.

5. NOAA National Centers for Environmental Information. Global historical climatology network hourly: documentation.

6. Durre I, Vose RS, Wuertz DB. Overview of the Integrated Global Radiosonde Archive. Journal of Climate. 2006;19:53–68. https://doi.org/10.1175/JCLI3594.1.

7. Durre I, Yin X, Vose RS, et al. Enhancing the data coverage in the Integrated Global Radiosonde Archive. Journal of Atmospheric and Oceanic Technology. 2018;35:1753–70. https://doi.org/10.1175/JTECH-D-17-0223.1.

8. NOAA National Centers for Environmental Information. Integrated global radiosonde archive: Product and sounding-format documentation.

9. Bryant C, Wheeler NR, Rubel F, French RH. Kgc: Koeppen-geiger climatic zones. 2017. https://doi.org/10.32614/CRAN.package.kgc.

10. Muñoz-Sabater J, Dutra E, Agustí-Panareda A, et al. ERA5-Land: A state-of-the-art global reanalysis dataset for land applications. Earth System Science Data. 2021;13:4349–83. https://doi.org/10.5194/essd-13-4349-2021.

11. Nussbaumer R, Adamik P, Alerstam T, Andersson A, Arlettaz R, Bächler E, et al. Master GeoLocator data package. 2026. https://doi.org/10.5281/zenodo.18187093.

12. Nussbaumer R, Gravey M. GeoPressureR: Global positioning by atmospheric pressure. 2022. https://doi.org/10.5281/zenodo.7754457.